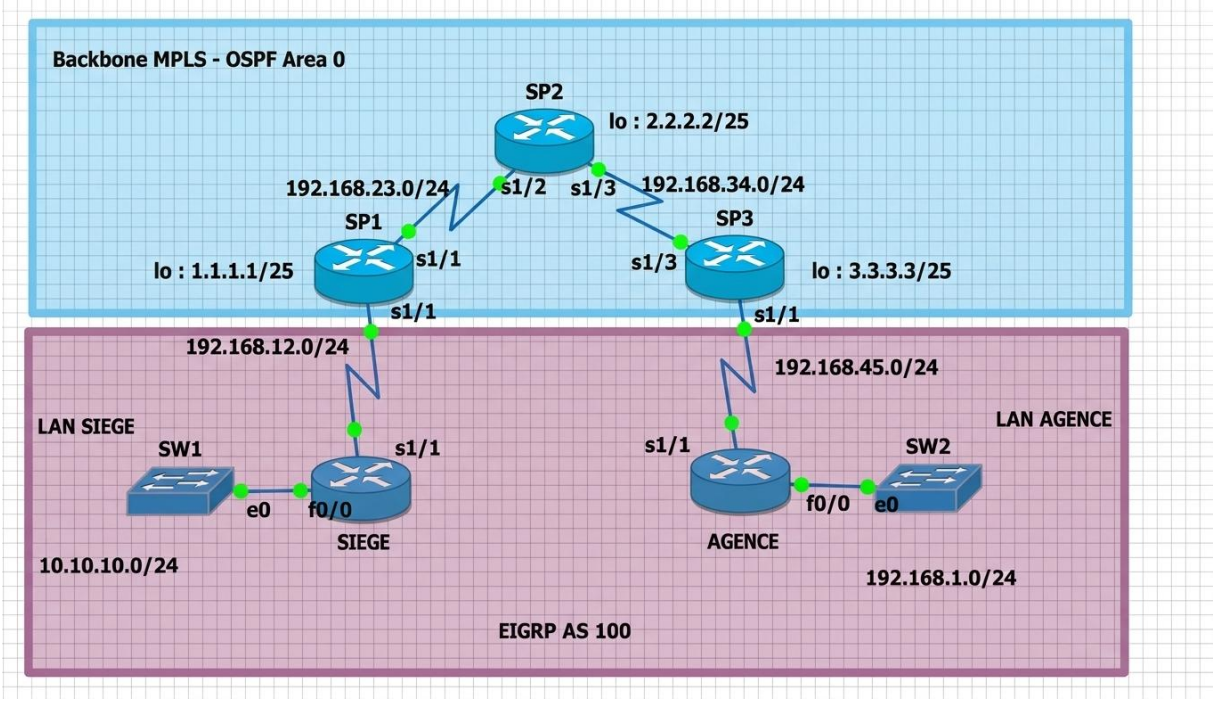
TP1 : Configuration d'un backbone MPLS

Malek Boughdiri RSI31



Plan d'adressage retenu

Lien	Réseau	Extrémité A	Extrémité B
SIEGE ↔ SP1	192.168.12.0/24	SIEGE s1/1 : .1	SP1 s1/1 : .2
SP1 ↔ SP2	192.168.23.0/24	SP1 s1/2 : .1	SP2 s1/2 : .2
SP2 ↔ SP3	192.168.34.0/24	SP2 s1/3 : .1	SP3 s1/3 : .2
SP3 ↔ AGENCE	192.168.45.0/24	SP3 s1/1 : .1	AGENCE s1/1 : .2
LAN SIEGE	10.10.10.0/24	SIEGE f0/0 : .1	—
LAN AGENCE	192.168.1.0/24	AGENCE f0/0 : .1	—

Etape 1 : Configuration de base des équipements réseau

SP1

```
hostname SP1
!
interface Loopback0
 ip address 1.1.1.1 255.255.255.128
!
interface Serial1/1
 ip address 192.168.12.2 255.255.255.0
 no shutdown
!
interface Serial1/2
 ip address 192.168.23.1 255.255.255.0
 no shutdown
```

SP2

```
hostname SP2
!
interface Loopback0
 ip address 2.2.2.2 255.255.255.128
!
interface Serial1/2
 ip address 192.168.23.2 255.255.255.0
 no shutdown
!
interface Serial1/3
 ip address 192.168.34.1 255.255.255.0
 no shutdown
```

SP3

```
hostname SP3
!
interface Loopback0
 ip address 3.3.3.3 255.255.255.128
!
interface Serial1/3
 ip address 192.168.34.2 255.255.255.0
 no shutdown
!
interface Serial1/1
 ip address 192.168.45.1 255.255.255.0
 no shutdown
```

SIEGE

```
hostname SIEGE
!
interface FastEthernet0/0
 ip address 10.10.10.1 255.255.255.0
 no shutdown
!
interface Serial1/1
 ip address 192.168.12.1 255.255.255.0
 no shutdown
```

AGENCE

```
hostname AGENCE
!
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
!
interface Serial1/1
 ip address 192.168.45.2 255.255.255.0
 no shutdown
```

Vérification demandée (connectivité point à point) :

```
SIEGE#ping 192.168.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
SIEGE#
```

```
AGENCE#ping 192.168.45.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
AGENCE#
```

Étape 2 : OSPF dans le réseau opérateur (SP1, SP2, SP3) :

SP1

```
router ospf 1
 router-id 1.1.1.1
 network 192.168.12.0 0.0.0.255 area 0
 network 192.168.23.0 0.0.0.255 area 0
 network 1.1.1.1 0.0.0.0 area 0
!
interface Loopback0
 ip ospf network point-to-point
```

SP2

```
router ospf 1
 router-id 2.2.2.2
 network 192.168.23.0 0.0.0.255 area 0
 network 192.168.34.0 0.0.0.255 area 0
 network 2.2.2.2 0.0.0.0 area 0
!
interface Loopback0
 ip ospf network point-to-point
```

SP3

```
router ospf 1
 router-id 3.3.3.3
 network 192.168.34.0 0.0.0.255 area 0
 network 192.168.45.0 0.0.0.255 area 0
 network 3.3.3.3 0.0.0.0 area 0
!
interface Loopback0
 ip ospf network point-to-point
```

Vérification de connectivité totale dans le domaine OSPF :

```
SP1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/  -        00:00:38    192.168.23.2    Serial1/2
```

```
SP1#show ip route ospf

     2.0.0.0/25 is subnetted, 1 subnets
O       2.2.2.2 [110/64] via 192.168.23.2, 00:12:03, Serial1/2
     3.0.0.0/25 is subnetted, 1 subnets
O       3.3.3.3 [110/128] via 192.168.23.2, 00:11:40, Serial1/2
     192.168.34.0/24 [110/128] via 192.168.23.2, 00:11:40, Serial1/2
O       192.168.34.0/24 is subnetted, 1 subnets
```

```
SP1#ping 3.3.3.3 source loopback0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
Packet sent with a source address of 1.1.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/6/8 ms
```

```
SP2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/  -        00:00:35    192.168.23.1    Serial1/2
3.3.3.3           1   FULL/  -        00:00:39    192.168.34.2    Serial1/3
```

```
SP2#show ip route ospf

     1.0.0.0/25 is subnetted, 1 subnets
O       1.1.1.1 [110/64] via 192.168.23.1, 00:12:10, Serial1/2
     3.0.0.0/25 is subnetted, 1 subnets
O       3.3.3.3 [110/64] via 192.168.34.2, 00:11:52, Serial1/3
O    192.168.12.0/24 [110/128] via 192.168.23.1, 00:12:10, Serial1/2
O    192.168.45.0/24 [110/128] via 192.168.34.2, 00:11:52, Serial1/3
```

```
SP2#ping 3.3.3.3 source loopback0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
Packet sent with a source address of 2.2.2.2
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/4 ms
```

```
SP3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/  -        00:00:37    192.168.34.1    Serial1/3
```

```
SP3#show ip route ospf

     1.0.0.0/25 is subnetted, 1 subnets
O       1.1.1.1 [110/128] via 192.168.34.1, 00:11:55, Serial1/3
     2.0.0.0/25 is subnetted, 1 subnets
O       2.2.2.2 [110/64] via 192.168.34.1, 00:11:55, Serial1/3
O    192.168.12.0/24 [110/192] via 192.168.34.1, 00:11:55, Serial1/3
O    192.168.23.0/24 [110/128] via 192.168.34.1, 00:11:55, Serial1/3
```

```
SP3#ping 3.3.3.3 source loopback0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
Packet sent with a source address of 3.3.3.3
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/6/9 ms
```

Vérifier les connectivités point à point entre AGENCE-SP3 et SIEGE-SPI

Lien SIEGE ↔ SP1 (192.168.12.0/24)

Depuis SIEGE

```
SIEGE#show ip interface brief
Interface              IP-Address      OK? Method Status                Protoc
FastEthernet0/0        10.10.10.1      YES manual up                    up
Serial1/1              192.168.12.1    YES manual up                    up
```

```
SIEGE#ping 192.168.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

```
SIEGE#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
SP1              Ser 1/1           178              R      7200      Ser 1/1
```

Depuis SP1

```
SP1#show ip interface brief
Interface              IP-Address      OK? Method Status                Proto
Serial1/1              192.168.12.2    YES manual up                    up
Serial1/2              192.168.23.1    YES manual up                    up
Loopback0              1.1.1.1         YES manual up                    up
```

```
SP1#ping 192.168.12.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

Lien AGENCE ↔ SP3 (192.168.45.0/24)

Depuis AGENCE

```
AGENCE#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
Serial1/1              192.168.45.2    YES manual up                    up
```

```
AGENCE#ping 192.168.45.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

```
AGENCE#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
SP3              Ser 1/1           176             R      7200      Ser 1/1
```

Depuis SP3

```
SP3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
Serial1/1              192.168.45.1    YES manual up                    up
Serial1/3              192.168.34.2    YES manual up                    up
Loopback0              3.3.3.3         YES manual up                    up
```

```
SP3#ping 192.168.45.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

Étape 3 : EIGRP dans le réseau client (SIEGE, AGENCE)

SIEGE

```
router eigrp 100
 network 10.10.10.0 0.0.0.255
 network 192.168.12.0 0.0.0.255
 no auto-summary
```

AGENCE

```
router eigrp 100
 network 192.168.1.0 0.0.0.255
 network 192.168.45.0 0.0.0.255
 no auto-summary
```